Chapter 12 Notes

Section 0

Notes

Note that when the relation $x = y$ is used (for instnace in P_1), this is a substitute for the formula

$$\forall z(z \in x \equiv z \in y)$$

Section 1

Exercise 1.1

We show that for finite α, $L_\alpha = R_\alpha$.

Proof. By induction. $L_0 = R_0$ by definition. Suppose $L_n = R_n$ where n is finite. By C_0, L_n is transitive (and thus also extensional), and hence by P_3, $\mathcal{F}(L_n) = \mathcal{P}(L_n) = \mathcal{P}(R_n) = R_{n+1}$, therefore completing the induction. □

Exercise 1.2

By Exercise 1.1, for finite α, $L_\alpha = R_\alpha$. Then it obviously follows that $L_\omega = R_\omega$. However, $L_{\omega+1} \neq R_{\omega+1}$. For recall that, if a set A is infinite, then $\mathcal{F}(A) \preceq A$. Thus $L_{\omega+1} = \mathcal{F}(L_\omega) \preceq L_\omega$ (because L_ω is infinite). But $R_{\omega+1} = \mathcal{P}(R_\omega) \succ R_\omega$ ($\succ L_\omega$) by Cantor's Theorem. Hence $L_{\omega+1} \neq R_{\omega+1}$, and $R_{\omega+1} \succ L_{\omega+1}$.